MATH102BKK · HARBOUR.SPACE BANGKOK

Homework 1 — Diagnostic

Session 1 · Course overview and school math refresher

Set
Mon 7 Sep 2026
Due
Tue 8 Sep 2026, start of class

HOW TO ANSWER **Every answer carries one sentence of justification.** A value on its own is not a solution — say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

Part A · Algebraic care

1 EXPONENTS

Simplify $\dfrac{27^{2/3} \cdot 3^{-2}}{9^{1/2}}$, stating which exponent law you use at each step.

2 RADICALS

Decide whether $\sqrt{x^2} = x$ holds for all real x. If not, give the correct identity and a value of x that breaks the false one.

3 IDENTITIES

Prove that $(a+b)^2 - (a-b)^2 = 4ab$ for all real a, b. State explicitly why your argument covers *all* real a, b and not just the ones you tried.

4 COMPLETING THE SQUARE

Write $2x^2 - 12x + 23$ in the form $a(x-h)^2 + k$ and read off its minimum value and where it occurs.

Part B · Absolute value and inequalities

5 ABSOLUTE VALUE

Solve $|3x+1| \geq 4$. Mark each step $\iff$ or $\implies$ and say why.

6 RATIONAL INEQUALITY

Find all real x with $\dfrac{2}{x-1} \leq 1$. Handle the sign of $x-1$ explicitly; do not multiply through by an expression of unknown sign.

7 AM–GM

Show $x^2 + y^2 \geq 2xy$ for all real x, y, and determine exactly when equality holds.

8 FINDING THE ERROR

A student writes: "Since $\frac{1}{x} < 1$, multiply both sides by x to get $1 < x$." Identify the false step, and give a value of x for which the premise is true but the conclusion is false.

Part C · Write like a mathematician

9 PRECISION

Each sentence below is sloppy. Rewrite it so it is unambiguous and true.

a. "The square root of a number is positive."

b. "If you square both sides of an inequality you get an equivalent inequality."

c. "$x^2 = 4$, so $x = 2$."

10 REFLECTION

In no more than eight sentences: pick one rule you were taught at school without a reason (for example the quadratic formula, or "you can't divide by zero"). State the rule precisely, then say what would have to be true for it to fail. You are not being asked to prove it.